

Commercialization of Cell & Gene Therapies in Canada: Current landscape, challenges and opportunities

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CGTs are redefining care for rare diseases, but Canada's commercialization, manufacturing, and access pathways lag behind peer jurisdictions.

About Us

- **PhD Candidate in Drug Development & Human Toxicology** at Queen's University under supervision of Dr. Louise Winn.
- Investigating **early molecular markers of disease risk during fetal development** following exposures to **environmental pollutants and pharmaceuticals**.
- Interested in the role of **policy and economics** in shaping health, both in **pharmaceutical development/healthcare** and **environmental protection**.





CGT Basics

- **Cell and gene therapies (CGTs)** offer potential for long-term or potentially even curative treatments.
- Especially relevant for rare genetic disease populations, and increasingly for oncology applications.

Canada's Drug Agency's Working Definition:

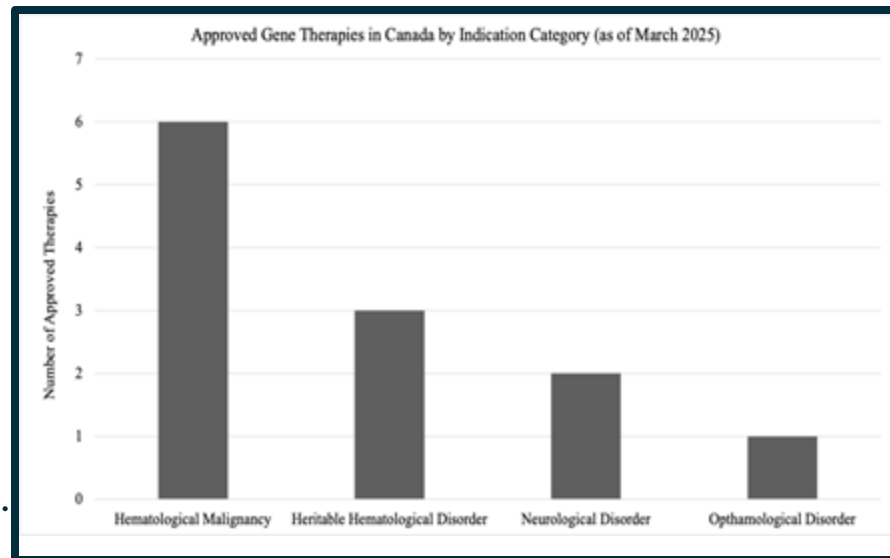
“A **set of strategies that modify the expression of an individual's genes** or repair abnormal genes involving the administration of a specific **nucleic acid** (i.e., DNA or RNA) via a viral or non-viral vector.

This includes immunotherapy involving **genetically modified T cells (e.g., CAR-T cell therapy)** and **regenerative medicine involving genetically modified cells or tissues**, including stem cells, and genome editing.”



The Canadian Context

- Canada has approved 13 CGTs since 2018 (first: Kymriah in 2018).
- 75% approved between 2020-2023, showing acceleration.
- Primarily target hematological malignancies (6 therapies) and heritable blood disorders (3 therapies).





The Canadian Context

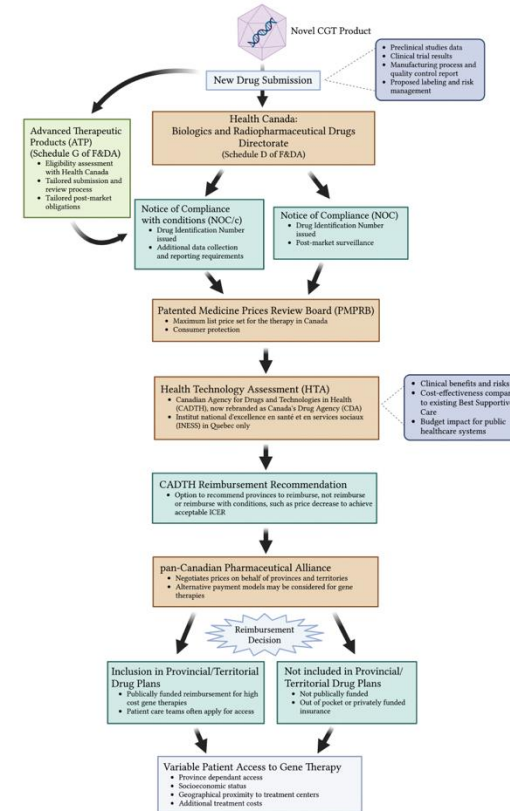
But we trail behind in number of approvals: US (44), EU (22) vs. Canada (13)

Critical gap hypothesis: Canada lacks the integrated infrastructure, policy frameworks and capital deployment needed to support domestic CGT commercialization.

What are the potential solutions?

The CGT Journey: from Lab to Patient

- A novel CGT product must navigate a complex and lengthy regulatory and reimbursement pathway to reach Canadian patients.
- Involving numerous governing body decisions surrounding safety, efficacy, pricing and cost-effectiveness.

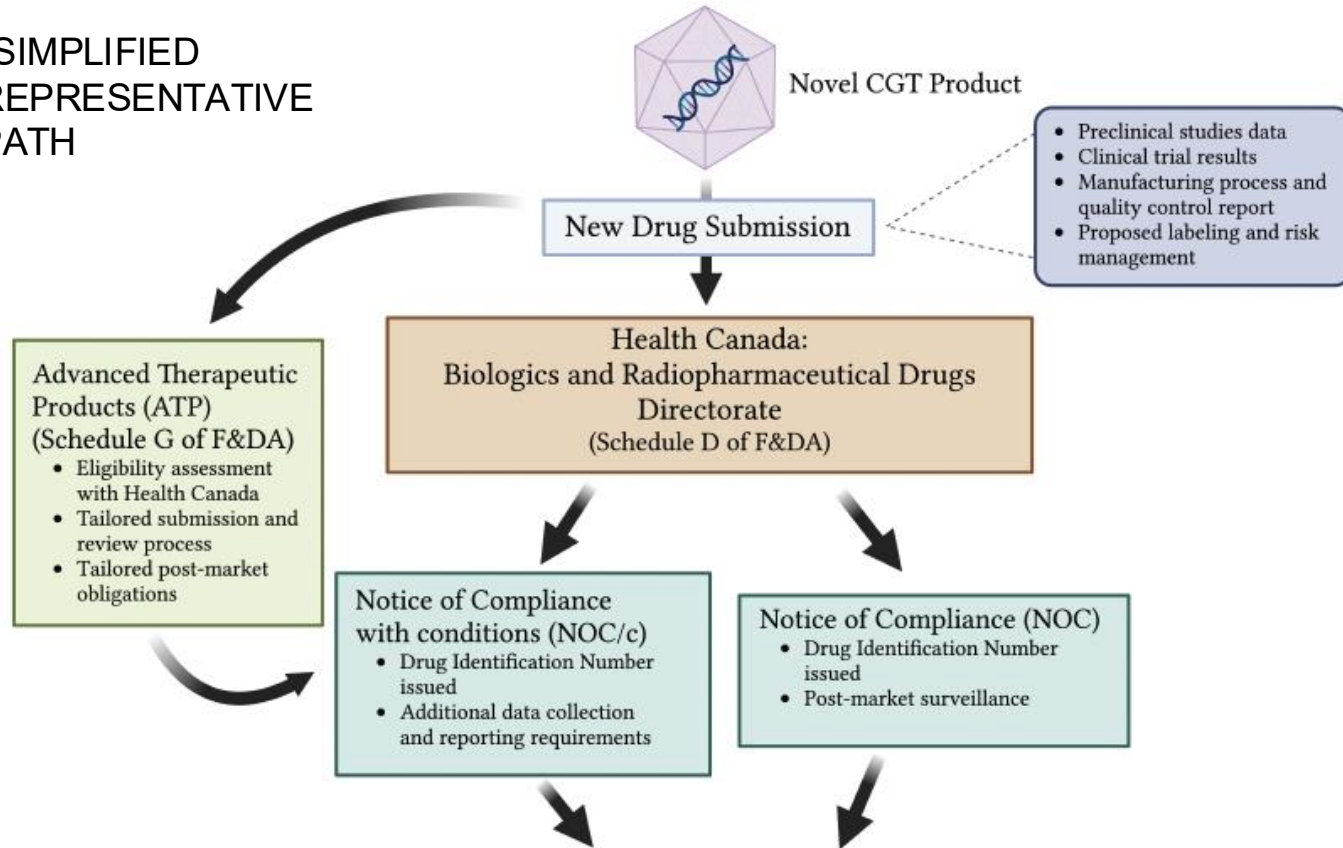


New
CGT

Patient
Access

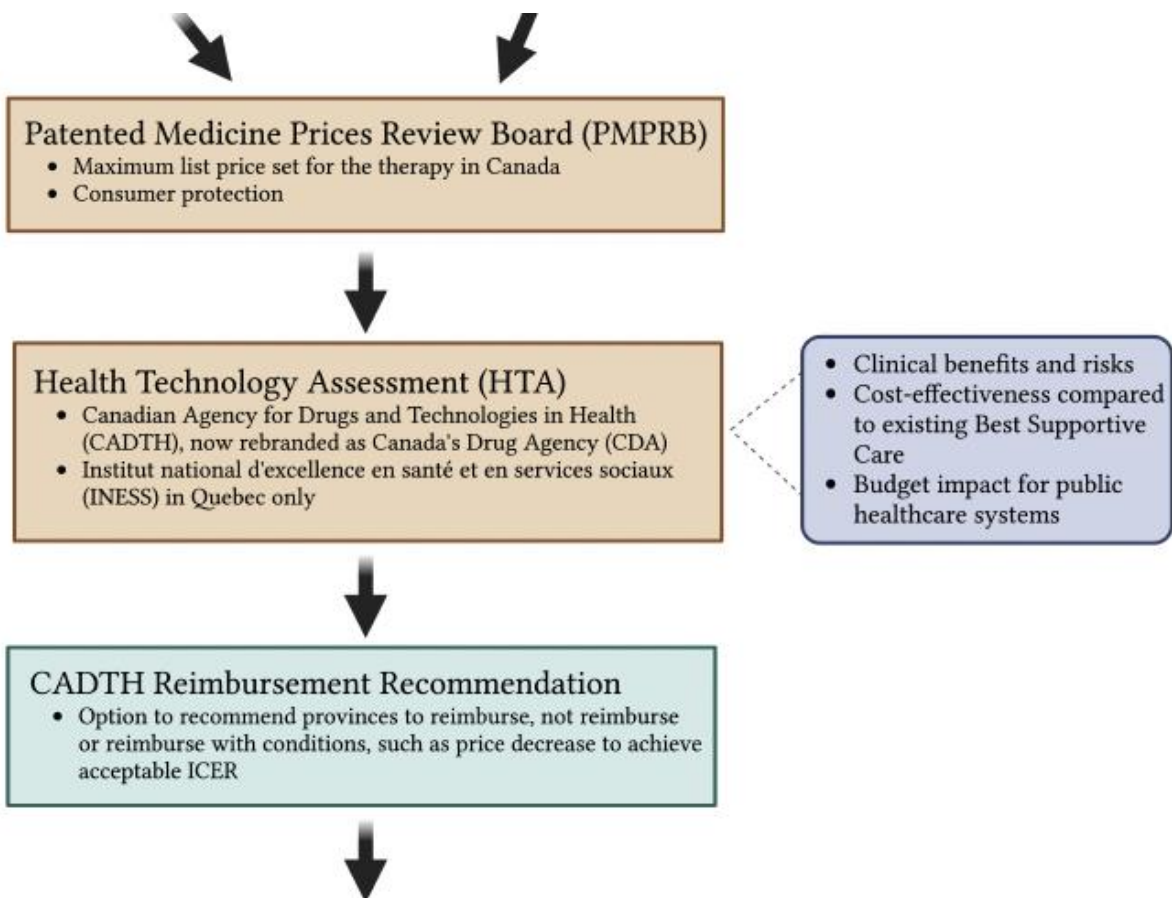
The CGT Journey: from Lab to Patient

*SIMPLIFIED
REPRESENTATIVE
PATH



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graph TD; A[ ] --> B[Patented Medicine Prices Review Board (PMPRB)]; A --> B; B --> C[Health Technology Assessment (HTA)]; C --> D[CADTH Reimbursement Recommendation];
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Patented Medicine Prices Review Board (PMPRB)

- Maximum list price set for the therapy in Canada
- Consumer protection

Health Technology Assessment (HTA)

- Canadian Agency for Drugs and Technologies in Health (CADTH), now rebranded as Canada's Drug Agency (CDA)
- Institut national d'excellence en santé et en services sociaux (INESS) in Quebec only

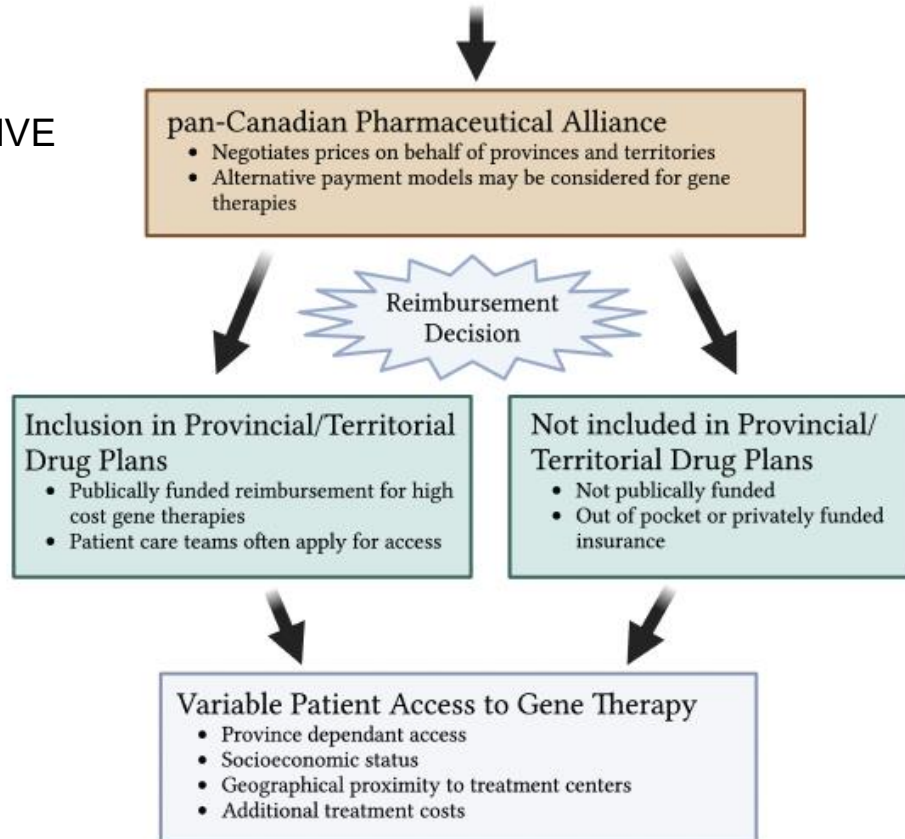
- Clinical benefits and risks
- Cost-effectiveness compared to existing Best Supportive Care
- Budget impact for public healthcare systems

CADTH Reimbursement Recommendation

- Option to recommend provinces to reimburse, not reimburse or reimburse with conditions, such as price decrease to achieve acceptable ICER

The CGT Journey: from Lab to Patient

*SIMPLIFIED
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PATH



→ Challenges & Opportunities across Four Key Domains

1

**Regulatory &
Policy**

2

**Manufacturing
& Development**

3

**Pricing &
Reimbursement**

4

**Access &
Equity**

- Challenges identified based on literature review
- Possible solutions pulled from international examples

→ Challenges & Opportunities across Four Key Domains

1. Regulatory & Policy



Regulatory & Policy

Challenge	Relevance to Canada	Possible Solutions?
No tailored CGT regulatory category (regulated as biologics)	Pathway design can be a competitive constraint - Reduced harmonization with tailored EU/US pathways.	Develop a distinct CGT category or tailored guidance, harmonized with EU/US submission pathways.



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No incentivized approval pathway	No orphan drug-style approval pathway; limits financial incentive vs. US and EU	Add financial incentives through regulatory alterations to the ATP pathway

→ Challenges & Opportunities
across Four Key Domains

2. Manufacturing & Development

→ Manufacturing & Development

Challenge	Relevance to Canada	Possible Solutions?
Domestic manufacturing infrastructure is nascent.	Canada's development & manufacturing capability of CGTs is limited (especially for small/mid-sized pharma).	Enhance public-private partnerships to support end-to-end innovation through to GMP manufacturing. • BioHubNet • OmniaBio • CCRM

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High manufacturing & distribution costs.	Personalized nature of CGTs creates complex logistics and limited scalability	Automated platforms and decentralized point-of-care manufacturing systems

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High manufacturing & distribution costs	Personalized nature of CGTs creates complex logistics and limited scalability	Automated platforms and decentralized point-of-care manufacturing systems
Limited domestic innovation → commercialization pipeline	Harder for Canadian SMEs to take CGTs from lab to market; drives foreign acquisition of innovations before domestic maturation	Scale existing models to fill capital gaps (CCRM, InvestOntario); establish joint-ventures and venture arms for commercialization-phase funding.

→ Challenges & Opportunities
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3. Pricing & Reimbursement



Pricing & Reimbursement

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Limitations of the traditional HTA process	Traditional ICER-based HTA fails to capture broader CGT value (curative potential, family burden, societal productivity)	Incorporate societal and patient-value assessment into HTAs

→ Challenges & Opportunities
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4. Access & Equity



Access & Equity

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Geographic and financial barriers to access	Rural patients face greater challenges in accessing treatment; inconsistent public & private coverage = differences in ability to pay	Leverage innovative insurance models (stop-loss?); travel assistance agreements; point-of-care manufacturing



Conclusions & Future Directions

Recent momentum:

- CAR-T is shifting CGTs from ultra-rare diseases to larger oncology populations
- First CRISPR approval = potential new applications.

Core challenge: Canada's CGT commercialization still lags peer jurisdictions due to interconnected hurdles across **regulation, manufacturing, reimbursement, and equity.**

Path forward: Learn from what is working internationally, then adapt solutions to the Canadian context.

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Strategic opportunity: Global CGT investment down 33% since 2021 → window for Canadian policy intervention and domestic investment during market consolidation time and position for next upswing.



Thank you for listening!

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Lihua Xue

Lauren Brown

Megan Cull

Perri Grant

Delaine Pereira

Katryna Seabrook

Holly MacKay



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